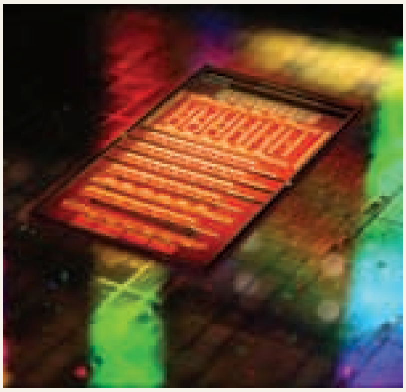
System integration Silicon Photonics



The basic idea behind Silicon photonics is to replace electricity as the medium of data transfer components on a board, with light. As fiber optics have revolutionised our long distance communication, efforts are on now to bring that efficiency over very short distances. This will be especially important as more and more components will be integrated on the same chip. It will rid us of the bottleneck that is set to develop during inter-component communication as the processing power increases. The best part is that these Silicon photonic devices will use the same fabrication technology that we have been using till now. Thus, there will be no loss of density when shifting to these Silicon photonic devices.

The PC is still here

After many predictions over the years of its imminent demise, the PC is still alive and kicking. Check out the modern day PC in its new avatar

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Human beings love to predict the demise of things. Besides their own race, I have been the second most popular item on the 'things-that'll-be-in-hell-soon' list. But Rapture '11 and Doomsday '12 came and went by; the human race is still here. So am I.

There are many reasons I still exist – the primary one being the human race's love for continually evolving bigger better things that the smaller mobiles and tablets have never been able to catch up with. When I was playing 720p video, these devices were on VGA. When I was pwning 1080p, they barely started chewing on the 720p. Even when they started catching up, the people who developed these high quality videos and games couldn't do it on mobiles, or even tablets (sorry iMovie, you're amateur). 1080p to 3D, and then on to holographic videos – people couldn't get enough of pixels, and so I lived on. RayWJ has been using me for the past 2 decades. I am indispensable!

The insides have been changing at a fast pace. I have a 120-core processor, and they are doubling that number in a couple of years. Those who are interested to know – yes, the damned Moore's Law is still holding: The beast

NVRAM

One thing that irks almost everyone is the booting up time of sophisticated computer devices – well, at least we hate it. This is because while you are using the computer, your 'session' is stored in the RAM, which is a volatile memory, i.e. it loses all its data as soon as it stops receiving power. In the future however, we are set to get what is called non-volatile RAM. In fact, a type of NVRAM is already available: SSDs! These flash memory chips are a type of NVRAM because they give you random access any of the data stored anywhere on drive can be accessed in equal time. This gives them excellent read times, but they don't have good enough write times, or reliability to replace your RAM. By reliability, we mean that this flash memory has a limited number of write/erase cycles. FeRAM (ferroelectric RAM) and MRAM (Magnetoresistive RAM) are two of the most promising NVRAM technologies coming up. The milipede memory technique developed by IBM is another innovative solution, which uses the age-old punch card technique, but at the nanometre scale!